

EXPERIMENT 4

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Experiment 4 -- To study and implement advanced SQL queries using Joins, Corelated Query's and Conditional statement.

Aim : To analyze and implement correlated subqueries, aggregate functions, and joins in SQL to retrieve employees whose salaries rank among the top three distinct salary values within their respective departments.

Question :4 Part A (i):

You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

```
Select rankings, country
FROM (
  SELECT rankings, country, (
    CASE
      WHEN country = 'India' THEN 1
      WHEN country = 'Sri Lanka' THEN 2
      ELSE 3
    END
  ) as ranks
FROM tbl_happiness
ORDER BY ranks, rankings
)
```

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	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	1	Finland
4	2	Denmark
5	3	Iceland
6	4	Israel
7	5	Netherlands
8	6	Sweden
9	7	Norway

Question: Part A (ii)

Make all the rankings in descending order except **India** and **Sri Lanka** SQL

```

Select rankings, country
FROM (
  SELECT rankings, country, (
    CASE
      WHEN country = 'India' THEN 1
      WHEN country = 'Sri Lanka' THEN 2
      ELSE 3
    END
  ) as ranks
FROM tbl_happiness
ORDER BY ranks ASC, rankings DESC
)

```

	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	7	Norway
4	6	Sweden
5	5	Netherlands
6	4	Israel
7	3	Iceland
8	2	Denmark
9	1	Finland

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Question: Part B

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

```
select m.employee_id, m.name ,count(*), sum(e.age)/count(*) as
average_age from employees as e JOIN employees as m on e.reports_to =
m.employee_id group by m.employee_id, m.name
```

	employee_id [PK] integer	name character varying (50)	reports_count bigint	average_age numeric
1	9	Hercy	2	39

Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand** the concept and implementation of correlated subqueries in SQL for solving row-wise comparison problems.
2. **Apply** aggregate functions such as `COUNT()` and `DISTINCT`, along with ranking logic, to retrieve the top three distinct salary values within each department.
3. **Implement** SQL joins and filtering techniques to combine data from multiple tables and generate accurate department-wise employee reports.
4. **Analyze** and optimize SQL queries involving grouped data, correlated subqueries, aggregate functions, and duplicate value handling to solve complex database problems effectively.